NATIONAL UNIVERSITY OF SINGAPORE

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MA1522 Linear Algebra for Computing **Graded Quiz 3**

1. (1) $\text{span}\{\mathbf{0}\} = \{\mathbf{0}\}$.
(2) $\text{span}(\varnothing) = \{\mathbf{0}\}$.

Solution. (1) True. Any vector in $\text{span}\{\mathbf{0}\}$ has the form $c\mathbf{0} = \mathbf{0}$; so $\text{span}\{\mathbf{0}\} = \{\mathbf{0}\}$.
(2) True; see Pg 105 of Chapter 3.

2. Let S and T be subsets of $\mathbb{R}^n$ such that $S \subseteq T$.
(1) $S \subseteq \text{span}(S)$.
(2) $\text{span}(S) \subseteq \text{span}(T)$.
(3) $S \subseteq \text{span}(T)$.

Solution. (1) True. Suppose $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$. Then

$$\boldsymbol{v}_i = 0\boldsymbol{v}_1 + \cdots + 0\boldsymbol{v}_{i-1} + 1\boldsymbol{v}_i + 0\boldsymbol{v}_{i+1} + \cdots + 0\boldsymbol{v}_k$$

is a linear combination of vectors in S. Hence, $\boldsymbol{v}_i \in \text{span}(S)$ for all i. So $S \subseteq \text{span}(S)$.
(2) True. If $\boldsymbol{v} \in \text{span}(S)$, i.e., $\boldsymbol{v}$ is a linear combination of vectors in S, then $\boldsymbol{v}$ is clearly a linear combination of vectors in T (as $S \subseteq T$), i.e., $\boldsymbol{v} \in \text{span}(T)$. Hence, $\text{span}(S) \subseteq \text{span}(T)$.
(3) True. It is the combination of (1) and (2): $S \subseteq \text{span}(S) \subseteq \text{span}(T)$.

3. (1) $\{\mathbf{0}\}$ is a subspace.
(2) $\varnothing$ is a subspace.

Solution. (1) True. $\{\mathbf{0}\} = \text{span}\{\mathbf{0}\}$; so it is a subspace.
(2) False. Any subspace must contain the zero vector $\mathbf{0}$; so $\varnothing$ is not a subspace.

4. Let V be a non-empty subset of $\mathbb{R}^n$.
(1) If V is a subspace, then V is closed under scalar multiplication and vector addition.
(2) If V is closed under scalar multiplication and vector addition, then V is a subspace.

Solution. (1) True; see Pg 58 of Chapter 3.
(2) True. Since V is non-empty, there exists $\boldsymbol{v} \in V$. Then $0\boldsymbol{v} \in V$, since V is closed under scalar multiplication. But $0\boldsymbol{v} = \mathbf{0}$. Thus V is a subspace by Pg 58 of Chapter 3.

5. Let S and T be subsets of $\mathbb{R}^n$ such that $\text{span}(S) = \text{span}(T)$.
(1) $S = T$.

(2) $|S| = |T|$.

Solution. Let $S = \{(1,0)\}$ and $T = \{(1,0),(2,0)\}$. Then $\text{span}(S) = \text{span}(T)$, which represents the x-axis in the xy-plane.

(1) False. It is clear that $S \neq T$.

(2) False. $|S| = 1$ and $|T| = 2$.

6. Let V be a subspace of $\mathbb{R}^n$ and let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$.

(1) If $\boldsymbol{a} \in V$ and $\boldsymbol{b} \in V$, then $\boldsymbol{a} + \boldsymbol{b} \in V$.

(2) If $\boldsymbol{a} \notin V$ and $\boldsymbol{b} \notin V$, then $\boldsymbol{a} + \boldsymbol{b} \notin V$.

Solution. (1) True. A subspace is closed under vector addition.

(2) False. Let $V = \{\mathbf{0}\}$ be the zero vector subspace. For any nonzero vector $\boldsymbol{a} \in \mathbb{R}^n$, let $\boldsymbol{b} = -\boldsymbol{a}$, then $\boldsymbol{a} \notin V$ and $\boldsymbol{b} \notin V$ but $\boldsymbol{a} + \boldsymbol{b} = \mathbf{0} \in V$.

7. Let U and V be subspaces of $\mathbb{R}^n$.

(1) $U \cup V$ is a subspace of $\mathbb{R}^n$.

(2) $U \cap V$ is a subspace of $\mathbb{R}^n$.

Solution. (1) False. Let $U = \text{span}\{(1,0)\}$ and $V = \text{span}\{(0,1)\}$. Then

$$(1,0) \in U \cup V, \quad (0,1) \in U \cup V, \quad \text{but} \quad (1,0) + (0,1) = (1,1) \notin U \cup V.$$

Since $U \cup V$ is not closed under vector addition, it is not a subspace.

(2) True. $U \cap V$ contains $\mathbf{0}$ and is closed under vector addition and scalar multiplication since both U and V have these properties.

8. (1) $\{\mathbf{0}\}$ is linearly independent.

(2) $\varnothing$ is linearly independent.

Solution. (1) False. The equation $c\mathbf{0} = \mathbf{0}$ has infinitely many solutions $c \in \mathbb{R}$; so $\{\mathbf{0}\}$ is linearly dependent.

(2) True. If $\varnothing$ is linearly dependent, there would exist $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k \in \varnothing$ such that $c_1\boldsymbol{v}_1 + \cdots + c_k\boldsymbol{v}_k = \mathbf{0}$ for some $c_1, \ldots, c_k$ not all zero. This contradicts the definition of empty set $\varnothing$. Hence, $\varnothing$ is linearly independent.

9. Let S and T be non-empty finite subsets of $\mathbb{R}^n$ such that $S \subseteq T$.

(1) If S is linearly independent, then T is linearly independent.

(2) If T is linearly independent, then S is linearly independent.

Solution. (1) False. Let $S = \{(1,0)\}$ and $T = \{(1,0),(2,0)\}$. Then $S \subseteq T$, S is linearly independent but T is linearly dependent.

(2) True; see Pg 92 of Chapter 3.

10. Let U and V be subspaces of $\mathbb{R}^n$ such that $U \subseteq V$.

(1) If B is a basis for V, then $B \cap U$ is a basis for U.
(2) There exists a basis C for V such that $C \cap U$ is a basis for U.

Solution. (1) False. Let $U = \text{span}\{(1,0)\}$ and $V = \mathbb{R}^2$. Then $B = \{(1,1),(0,1)\}$ is a basis for V but $B \cap U = \varnothing$ is not a basis for U.
(2) True. Let C' be a basis for U. Extend C' to a basis C for V by repeatedly applying Theorem in Pg 91 of Chapter 3. Then $C \cap U = C'$ is a basis for U.